# MOEA/D-DAP: Angle-Guided Adaptive Preference Decomposition for More Informative ROI Search


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***Abstract*—Preference-based multi-objective optimization should not only drive the population into a region of interest (ROI), but also return solutions in a form that genuinely helps decision makers. Existing decomposition-based methods usually distribute solutions almost uniformly inside the ROI. This is convenient for search, yet weak for interpretation: points near the preference point should be denser, and neighboring directions should not repeatedly encode nearly identical trade-off information. To address this mismatch, we propose MOEA/D-DAP, an angle-guided adaptive preference decomposition method built on MOEA/D. The method combines three ideas: a centered ROI construction that keeps the projected preference point at the geometric center of the preferred region, an adaptive density rule that allocates more reference vectors near the preferred compromise, and an angle-guided perturbation strategy that steers neighboring vectors toward under-covered sectors. Experiments on ZDT, DTLZ, and WFG benchmarks with 2 to 10 objectives show that MOEA/D-DAP achieves consistently stronger convergence and preference-aware distribution than representative baselines.**



***Index Terms*—Preference optimization, MOEA/D, decomposition, region of interest, many-objective optimization**


## I. Introduction

Many real-world engineering problems involve multiple conflicting objectives, such as performance, cost, and reliability. These problems are typically formulated as multi-objective optimization problems (MOPs), where evolutionary multi-objective optimization (EMO) algorithms are widely used to approximate Pareto-optimal trade-offs. Recent studies have further improved the scalability and practicality of EMO in many-objective and real-world settings [1]–[3]. However, returning the entire Pareto front often produces a large solution set that is difficult for decision makers to analyze, motivating the use of preference-based optimization.

In many real decision problems, decision makers (DMs) do not need the entire Pareto front. They care about a narrow area around a preferred trade-off, commonly expressed by a preference point (PP) or a reference point. Preference-based evolutionary multi-objective optimization therefore aims to search only the region of interest (ROI), which can greatly reduce computational waste and make the final solution set easier to inspect.

Among existing methods, decomposition-based algorithms are attractive because they usually scale better than dominance-based methods in many-objective settings [4], [5]. Interactive and reference-direction-based preference methods have long shown that optimization quality depends not only on convergence, but also on how well the returned set reflects the DM's intended trade-off structure [6]. However, most ROI-based decomposition methods still inherit a uniform-search mindset: once an ROI is defined, the algorithm tends to fill it evenly. This is computationally simple but conceptually incomplete. From the DM's perspective, solutions close to the PP are usually more valuable than those near the ROI boundary. Meanwhile, several uniformly distributed points may lie on very similar directions and thus carry almost the same preference information.

This paper revisits preference decomposition from a simple question: *how should a limited number of solutions be arranged so that they are easiest for the DM to understand and choose from?*

Instead of redesigning the whole EMO framework, we redesign only the reference-vector generation stage in MOEA/D. The resulting method, MOEA/D-DAP, has three contributions:

1) **Centered ROI construction:** the projected PP is always located at the geometric center of the generated ROI, avoiding truncation when the PP lies near the simplex boundary.
2) **Adaptive density control:** reference vectors are denser near the PP and sparser near the boundary, which better matches preference intensity.
3) **Angle-guided perturbation:** local directional disturbance reduces redundant sampling and increases preference-information coverage.

## II. Motivation

For an $m$-objective minimization problem

$$\min_{x\in\Omega} F(x) = \big(f_1(x), f_2(x), \ldots, f_m(x)\big), \tag{1}$$

MOEA/D solves scalar subproblems via the Tchebycheff aggregation

$$g^{\text{tch}}(x \mid \lambda, z^*) = \max_{1\le i\le m} \lambda_i \, |f_i(x) - z_i^*|, \tag{2}$$

where $\lambda$ is the reference vector and $z^*$ is the ideal point [4]. Hence, preference quality largely depends on how reference vectors are placed.

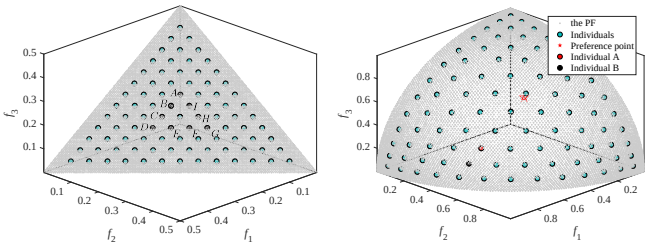



Fig. 1: Illustration of the uniform-distribution issue in MOEA/D-STM. Many points are evenly placed in the ROI, but several nearby points still encode highly similar preference directions.

Fig. 1 visualizes the first weakness of existing decomposition-based preference search. Although the solutions appear orderly, many of them are concentrated on similar rays from the PP. This creates redundant preference information while leaving other informative directions relatively under-explored. The issue becomes more serious in many-objective settings because the objective space expands rapidly and the cost of wasting reference vectors becomes higher.

A second weakness is that conventional preferred-region construction is boundary-sensitive. When the PP is close to the simplex boundary, the generated ROI can be skewed or partially truncated. As a result, the region searched by the algorithm is no longer centered on the information actually provided by the DM. Adaptive weight-vector strategies can partially reshape density [7], but they still do not directly solve the coupled problem of ROI centering, center-dense allocation, and directional redundancy. These observations motivate a reference-generation strategy that is both *preference-shaped* and *geometry-aware*: it should place more vectors near the PP while also improving angular coverage.

## III. MOEA/D-DAP

### A. Centered and Adaptive Reference Generation

Let

$$q = \left(\frac{1}{m}, \frac{1}{m}, \ldots, \frac{1}{m}\right) \tag{3}$$

be the center of the unit simplex. Given a PP $p = (p_1, \ldots, p_m)$, its projection onto the simplex is

$$p' = \frac{p}{\sum_{i=1}^{m} p_i}. \tag{4}$$

Rather than generating vectors directly around $p'$, MOEA/D-DAP first builds a temporary ROI around $q$ and then shifts it to $p'$. This guarantees that the final preferred region remains centered even for extreme PP locations.

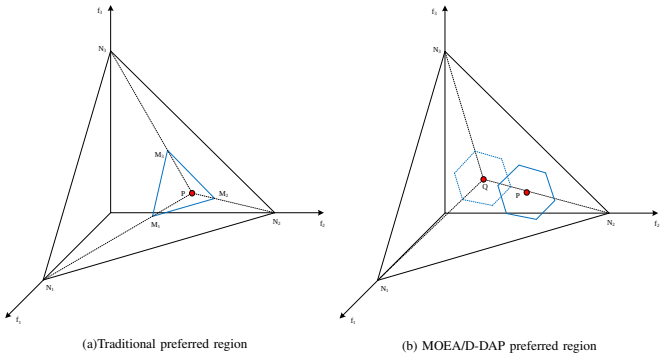



Fig. 2: Preferred-region construction. MOEA/D-DAP keeps the projected preference point at the center of the ROI instead of letting the region be pulled toward the boundary.

As shown in Fig. 2, the centered construction changes the logic of preference search. Instead of using the projected PP as an extreme anchor of the ROI, MOEA/D-DAP first creates a geometrically regular preferred region and then translates it so that the final ROI revolves around the projected PP. This avoids information loss near the simplex boundary and makes the returned set easier to interpret.

Suppose each boundary direction contains $n$ radial layers. For layer $\ell$, the radius is defined by

$$r_\ell = \delta^\ell r_0, \quad \ell = 0, 1, \ldots, n-1, \tag{5}$$

where $r_0$ is the boundary radius and

$$\delta = \beta^{\frac{1}{n-1}}, \qquad \beta = \frac{1}{\sigma n}. \tag{6}$$

Following the full manuscript, $\sigma = 2$. Since $0 < \delta < 1$, the layers contract toward the center, so more reference points are allocated near the PP. This directly implements the idea that preference intensity should be center-dense rather than uniform.

To avoid radial redundancy, each temporary reference vector is further perturbed by local angle information:

$$w_{\ell,j} = q + r_\ell d_j + \rho_{\ell,j} s_j, \tag{7}$$

where $d_j$ is a normalized boundary direction, $s_j$ is a shift direction computed from neighboring geometry, and $\rho_{\ell,j} \in [0, \beta]$ controls the perturbation magnitude. The first term ensures convergence toward the ROI, while the perturbation term pushes some vectors into blank angular sectors. For many-objective settings, this geometric organization is related in spirit to clustering ideas used to separate similar directional information [8]. Finally, the temporary vectors are translated to the DM's ROI by

$$w'_{\ell,j} = w_{\ell,j} + (p' - q). \tag{8}$$

Fig. 3 shows the geometric logic behind the method. In 2-D, reference points are contracted from the boundary toward the center and then shifted to the final ROI. In 3-D, angle

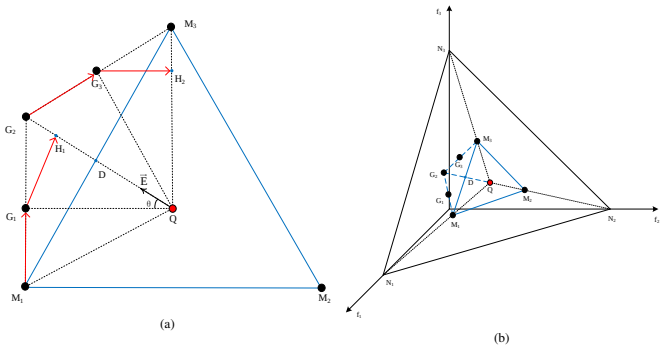



Fig. 3: Reference-generation mechanism in 2-D and 3-D spaces. The figure follows the original manuscript and illustrates temporary ROI construction, translation to the final ROI, and angle-guided perturbation.

TABLE I: Behavioral comparison of representative preference-search mechanisms.

| Method type | ROI centering | Density near PP | Boundary robustness | Direction coverage |
|---|---|---|---|---|
| Dominance-based | indirect | unstable | weak | ranking-dependent |
| Uniform decomposition | limited | almost uniform | weak | often redundant |
| Adaptive weights [7] | partial | center-biased | moderate | partly improved |
| **MOEA/D-DAP** | **explicit** | **center-dense** | **strong** | **angle-aware** |

decomposition provides a structured way to insert additional points between neighboring directions, while the perturbation term avoids repeatedly sampling the same preference ray. This mechanism is particularly helpful when the population budget is limited, because each reference vector should encode a clearly different trade-off pattern.

MOEA/D-DAP also naturally supports multiple ROIs. If the DM provides several preference points together with corresponding region sizes and solution budgets, the same centered-construction rule can be applied to each ROI independently, and the final reference-vector set is simply the union of all local sets. This makes the method practical when the DM is uncertain about a single exact compromise and wants several alternative preference neighborhoods to be explored simultaneously.

The design is simple but useful. **Centered ROI** ensures that the DM's preference is represented in the right location. **Adaptive density** gives the DM more alternatives near the most relevant compromise. **Angle guidance** reduces near-duplicate trade-offs and expands informative direction coverage. Since the subsequent search, mating, neighborhood update, and elitist replacement remain those of MOEA/D, MOEA/D-DAP can be integrated into existing decomposition frameworks with minimal code changes.

## IV. Experimental Study

### A. Setup

We compare MOEA/D-DAP with MOEA/D-STM [5], MOEA/D-PRE [9], g-NSGAII [10], r-NSGAII [11], and p-NSGAII. Test suites include ZDT, DTLZ, and WFG problems with 2, 3, 5, 8, and 10 objectives. Following the full manuscript, all algorithms use SBX crossover ($p_c = 1$, distribution index 10), polynomial mutation ($p_m = 0.1$, distribution index 20), and the same preferred-region size $\varepsilon = 0.1$. Each setting is executed 30 times in PlatEMO [12]. GD is used for convergence, while R-IGD and R-HV evaluate preference-aware quality.

TABLE II: Selected GD results on representative low-dimensional problems (mean values; feasible PP).

| Problem | Best baseline | MOEA/D-DAP | Gain |
|---|---|---|---|
| ZDT1 | 4.26e-6 (MOEA/D-STM) | **3.89e-6** | 1.10× |
| ZDT2 | 6.44e-6 (g-NSGAII) | **5.43e-6** | 1.19× |
| ZDT3 | 3.03e-5 (g-NSGAII) | **1.14e-5** | 2.65× |
| ZDT4 | 1.14e-3 (p-NSGAII) | **6.30e-4** | 1.81× |
| DTLZ1 | 1.63e-1 (MOEA/D-STM) | **4.85e-2** | 3.35× |
| DTLZ3 | 5.45e-1 (MOEA/D-STM) | **7.87e-2** | 6.93× |
| DTLZ6 | 1.75e-4 (MOEA/D-STM) | **6.16e-6** | 28.39× |
| WFG2 | 1.34e-2 (g-NSGAII) | **2.48e-3** | 5.37× |
| WFG6 | 1.14e-2 (g-NSGAII) | **6.93e-3** | 1.64× |
| WFG9 | 5.08e-3 (g-NSGAII) | **3.18e-3** | 1.60× |

TABLE III: Selected R-metric results on many-objective DTLZ problems.

| Problem | $M$ | Metric | Best baseline | MOEA/D-DAP | Gain |
|---|---|---|---|---|---|
| DTLZ2 | 8 | R-HV | 2.64e-1 (p-NSGAII) | **7.92e-1** | 3.01× |
| DTLZ4 | 8 | R-HV | 4.63e-1 (p-NSGAII) | **1.07e+0** | 2.32× |
| DTLZ4 | 10 | R-IGD | 8.00e-1 (g-NSGAII) | **1.06e-1** | 7.55× |
| DTLZ5 | 8 | R-HV | 9.76e-6 (g-NSGAII) | **1.81e-5** | 1.85× |
| DTLZ5 | 10 | R-IGD | 1.20e-1 (g-NSGAII) | **1.10e-1** | 1.09× |
| DTLZ6 | 5 | R-HV | 7.04e-1 (p-NSGAII) | **8.53e-1** | 1.21× |
| DTLZ6 | 8 | R-HV | 1.41e-5 (MOEA/D-STM) | **1.81e-5** | 1.28× |
| DTLZ6 | 10 | R-IGD | 4.41e-1 (MOEA/D-STM) | **1.03e-1** | 4.28× |

Table II shows that MOEA/D-DAP is especially effective on discontinuous, multimodal, and deceptive low-dimensional problems such as ZDT3, ZDT4, DTLZ3, and WFG2. These are exactly the scenarios where redundant vectors and a poorly centered ROI waste evaluations. On DTLZ6, for example, the GD value is improved from $1.75 \times 10^{-4}$ for the strongest baseline to $6.16 \times 10^{-6}$, which indicates that preference-shaped decomposition does not sacrifice convergence.

Table III further shows that the proposed method is particularly competitive on many-objective DTLZ problems when both convergence and preference-aware distribution are considered. The gains on DTLZ4 and DTLZ6 are especially notable, which agrees well with the geometric motivation of centering the ROI and redistributing vectors toward informative directions. This also supports the behavioral summary in Table I: the advantage of MOEA/D-DAP is not tied to a single metric, but comes from simultaneously improving location, density, and angular informativeness.

Fig. 4 explains *why* these gains occur. MOEA/D-STM tends to preserve nearly uniform layers, MOEA/D-PRE still leaves several sectors insufficiently explored, and dominance-based baselines become sparse or unstable around the ROI. In contrast, MOEA/D-DAP simultaneously preserves convergence toward the PP and increases angular coverage. This is exactly the desired behavior for preference optimization: the returned set is not only close to the DM's target, but also more informative to inspect.

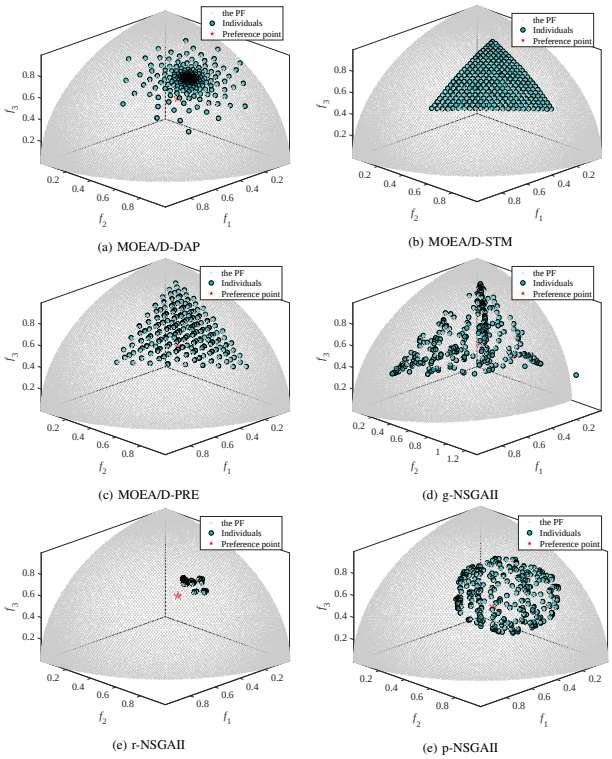



Fig. 4: Scatter plots of solutions on 3-D DTLZ2 with PP $(0.2, 0.3, 0.3)$. The subfigure naming style follows the full manuscript. MOEA/D-DAP yields a denser center and broader angular coverage, whereas the baselines are more uniform, clustered, or directionally incomplete.

A useful takeaway is that preference optimization should not be judged only by whether the population enters the ROI. Two algorithms may reach the same neighborhood of the PP, but the one that returns less redundant and more interpretable trade-offs is much more useful to the DM. This is why the improvement under R-metric is often even clearer than under GD in the full manuscript. In practical decision support, this difference matters because the DM rarely selects only the mathematically nearest point; instead, they compare several nearby alternatives and inspect how the trade-offs vary around the preferred compromise.

### *B. Discussion and Practical Interpretation*

Tables II–III and Fig. 4 together explain why MOEA/D-DAP is effective. GD improvements show that reshaping the reference vectors does not destabilize the underlying MOEA/D search, while the stronger R-IGD and R-HV values show a second, more important effect: the returned set is not only close to the PP, but also better organized around it. This distinction matters because a DM usually inspects several nearby candidates instead of selecting only the mathematically nearest point.

The three design components also play different roles. *Centered ROI* fixes the location problem, *adaptive density* fixes the local-resolution problem, and *angle-guided perturbation* fixes the redundancy problem. The method is therefore especially useful when the PP lies near the simplex boundary, when the population budget is limited, or when many objectives make repeated sampling of similar directions increasingly expensive. Since MOEA/D-DAP changes mainly the reference-vector generation stage, these gains are achieved with little disruption to the standard MOEA/D pipeline.

## V. Conclusion

This paper presents a conference version of MOEA/D-DAP, whose central message is that preference search should be center-dense, angle-aware, and geometrically well-centered. By combining centered ROI construction, adaptive density control, and angle-guided perturbation, the method returns solution sets that are easier for decision makers to interpret and compare. Results on ZDT, DTLZ, and WFG problems with up to 10 objectives confirm gains in both convergence and preference-aware distribution.

The method is lightweight because it mainly changes the reference-vector generation stage and leaves the MOEA/D search engine unchanged. Future work will study irregular Pareto-front geometries, adaptive perturbation magnitudes, and interactive settings with moving preference points.

## References


[1] J. Blank and K. Deb, "Pymoo: Multi-objective optimization in python," *Ieee access*, vol. 8, pp. 89 497–89 509, 2020.

[2] S. Sharma and V. Kumar, "A comprehensive review on multi-objective optimization techniques: Past, present and future: S. sharma, v. kumar," *Archives of Computational Methods in Engineering*, vol. 29, no. 7, pp. 5605–5633, 2022.

[3] H. Ishibuchi, Y. Nan, and L. M. Pang, "Performance evaluation of multi-objective evolutionary algorithms using artificial and real-world problems," in *International Conference on Evolutionary Multi-Criterion Optimization*. Springer, 2023, pp. 333–347.

[4] Q. Zhang and H. Li, "Moea/d: A multiobjective evolutionary algorithm based on decomposition," *IEEE Transactions on evolutionary computation*, vol. 11, no. 6, pp. 712–731, 2007.

[5] K. Li, R. Chen, G. Min, and X. Yao, "Integration of preferences in decomposition multiobjective optimization," *IEEE transactions on cybernetics*, vol. 48, no. 12, pp. 3359–3370, 2018.

[6] K. Deb and A. Kumar, "Interactive evolutionary multi-objective optimization and decision-making using reference direction method," in *Proceedings of the 9th annual conference on Genetic and evolutionary computation*, 2007, pp. 781–788.

[7] F. Wang, Y. Li, H. Zhang, T. Hu, and X.-L. Shen, "An adaptive weight vector guided evolutionary algorithm for preference-based multi-objective optimization," *Swarm and Evolutionary Computation*, vol. 49, pp. 220–233, 2019.

[8] R. Denysiuk, L. Costa, and I. Espírito Santo, "Clustering-based selection for evolutionary many-objective optimization," in *International conference on parallel problem solving from nature*. Springer, 2014, pp. 538–547.

[9] G. Yu, J. Zheng, R. Shen, and M. Li, "Decomposing the user-preference in multiobjective optimization," *Soft Computing*, vol. 20, no. 10, pp. 4005–4021, 2016.

[10] J. Molina, L. V. Santana, A. G. Hernández-Díaz, C. A. C. Coello, and R. Caballero, "g-dominance: Reference point based dominance for multiobjective metaheuristics," *European Journal of Operational Research*, vol. 197, no. 2, pp. 685–692, 2009.

[11] K. Deb and J. Sundar, "Reference point based multi-objective optimization using evolutionary algorithms," in *Proceedings of the 8th annual conference on Genetic and evolutionary computation*, 2006, pp. 635–642.

[12] Y. Tian, R. Cheng, X. Zhang, and Y. Jin, "Platemo: A matlab platform for evolutionary multi-objective optimization [educational forum]," *IEEE Computational Intelligence Magazine*, vol. 12, no. 4, pp. 73–87, 2017.